



National Institute of Business Management

Robotic Application Development

- Report -

(Greenhouse Automation)

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Introduction

Agriculture is undergoing a rapid technological transformation, with traditional methods moving towards Controlled Environment Agriculture (CEA). Among various methods, Greenhouse farming has become a vital technique utilizing specialized structures to accurately regulate environmental factors such as temperature, humidity, and sunlight. By optimizing these conditions, Greenhouses maximize crop yield, enhance quality, and extend growing seasons regardless of unavoidable weather patterns.

Problem Statement

Even though there are advantages of greenhouse farming, it remains restricted by two critical barriers:

- * **Cost Expensiveness:** Most commercial greenhouse systems rely on high-price-tag, industrial-grade equipment. The cost required for installation and maintenance makes these systems financially non affordable for small-scale farmers.
- * **Lack of Remote Accessibility:** Many existing industrial systems operate on localized control frameworks. Even though the system is autonomous the Farmer must be in premises to check the crop vitals.

The Solution: Affordable & Remote Capable Greenhouse Automation

To address these issues and cover this gap, we have introduced an **Affordable Remote Monitoring Capable Greenhouse Automation System**. This solution is specifically engineered to populate CEA technology by addressing the financial and logistical constraints of traditional systems. The system aims to deliver a low-cost, high-efficiency automation framework equipped with remote monitoring and control features, allowing small-scale farmers to manage their crops seamlessly from anywhere.

- * **Fully Automated Greenhouse System:** Arduino Uno R3 functions as the autonomous brain for the system
- * **Remote Monitoring Capabilities:** System is equipped with UART communication, a simple reverse proxy server can make the system accessible through internet as well.
- * **Affordability:** Fully featured system only costs around 5000 Rs.

Methodology

This project is implemented as a high-fidelity tabletop model, utilizing a water bottle reservoir and a PVC frame to simulate industrial greenhouse conditions. A critical architectural decision was the pivot from the **ESP32** to the **Arduino Uno R3** for the primary controller, in earlier stages before the project began with our decision it was to use ESP32 as the microcontroller, but the System being a low-powered one using an Arduino Uno became correct approach.

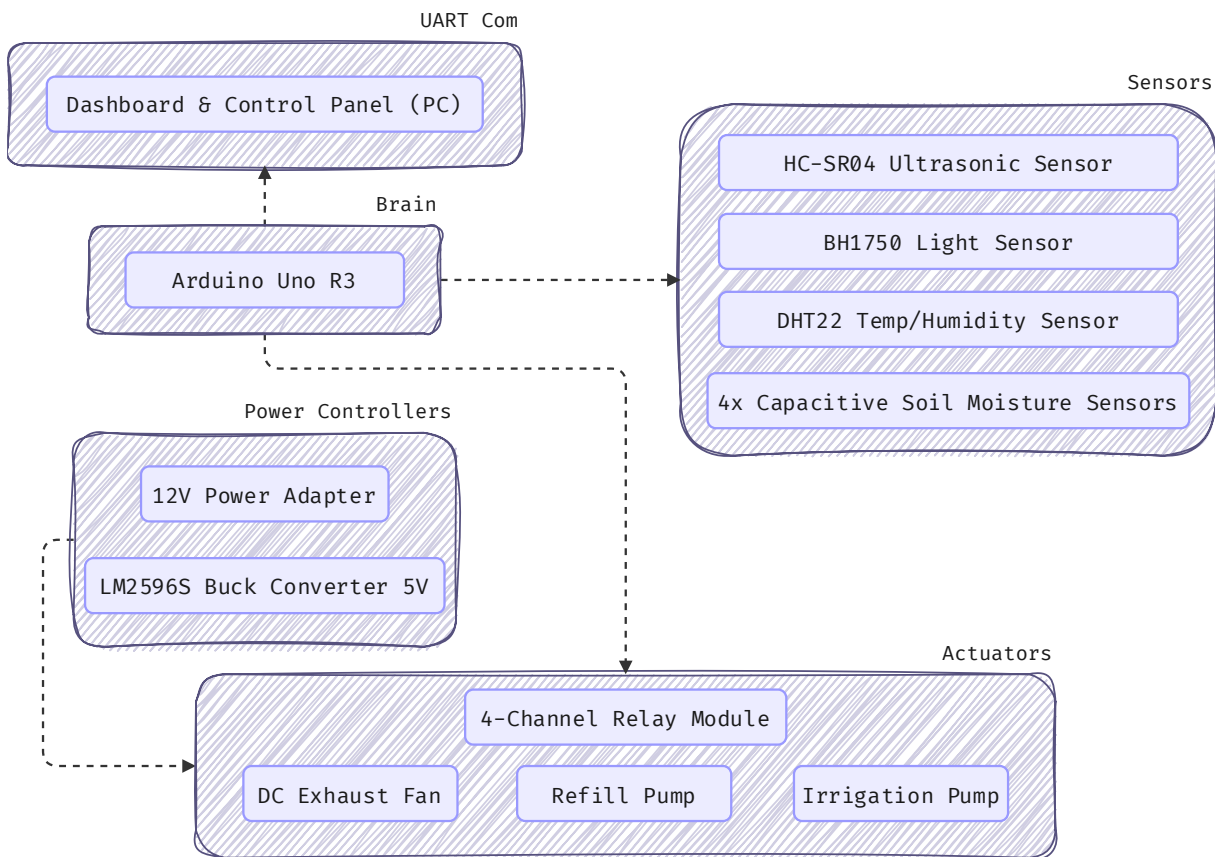


Figure 1 - System Design

Above is the confirmed system design before the implementation. We have isolated each of the systems to make it more maintainable and configurable, as you can see, we have isolated Power, Sensors, Serial Communications, Actuator and Microcontroller. We have designed the system as this because by making the design modular we can work on them each by each, and it also makes the installation process much easier.

Hardware Selection and Assembly

Below is how the system is assembled and components are connected to the Microcontroller.

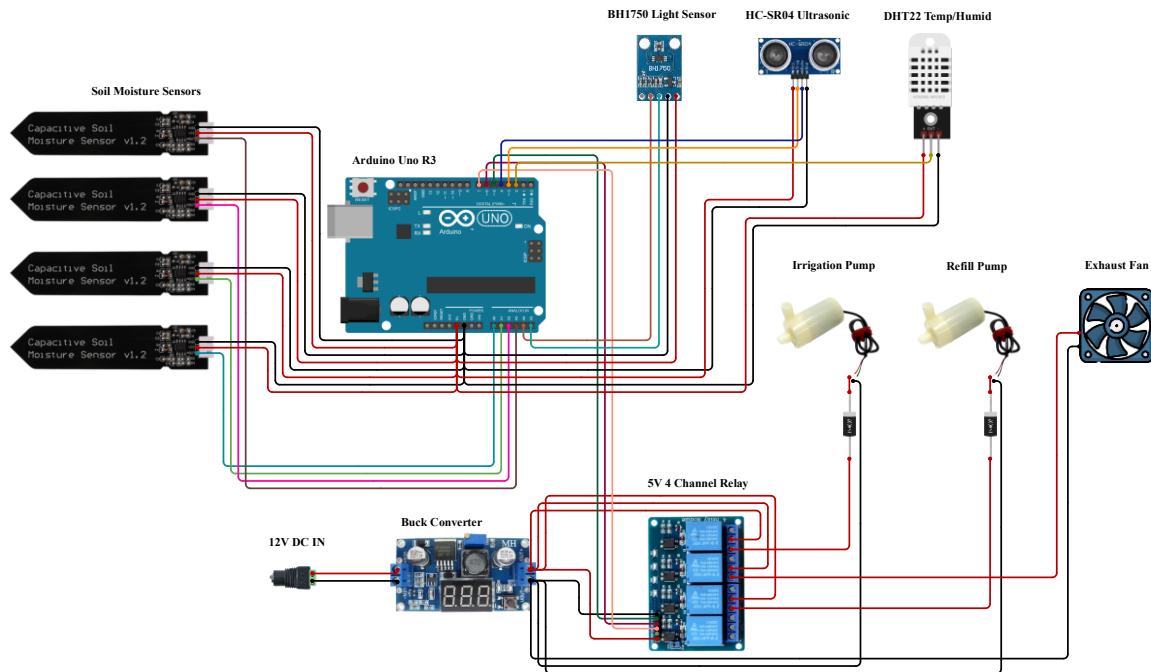


Figure 2 - Circuit Diagram

Hardware and Software Integration

The system follows strict "Middleman" architecture. By implementing a Hardware Layer, we ensure the **Arduino** is never burdened with high level logic. Arduino constantly monitors the system, executes the logic as needed while our custom-built **C#** middleman monitors the data and sends it to an interactive live dashboard for easy Greenhouse Monitoring

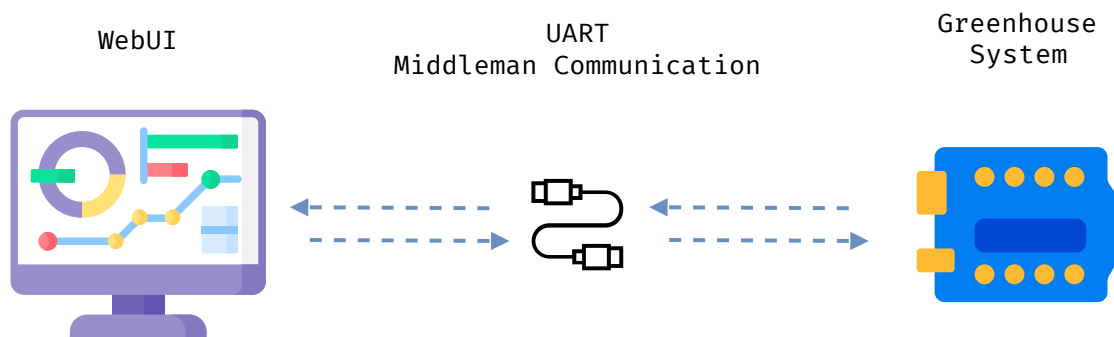


Figure 3 - UART Middleman Communication

System Design

Installing the stem was a challenge, we must identify how each component is placed and how they need to be connected. Distance to the sensors, Actuators and other relevant components matters in this scenario very much. And for some sensors it must be installed in specific way to measure data, once the circuit design is complete, we came up with the system design. Which covers all the scenarios and how the system needs to be installed in a professional manner.

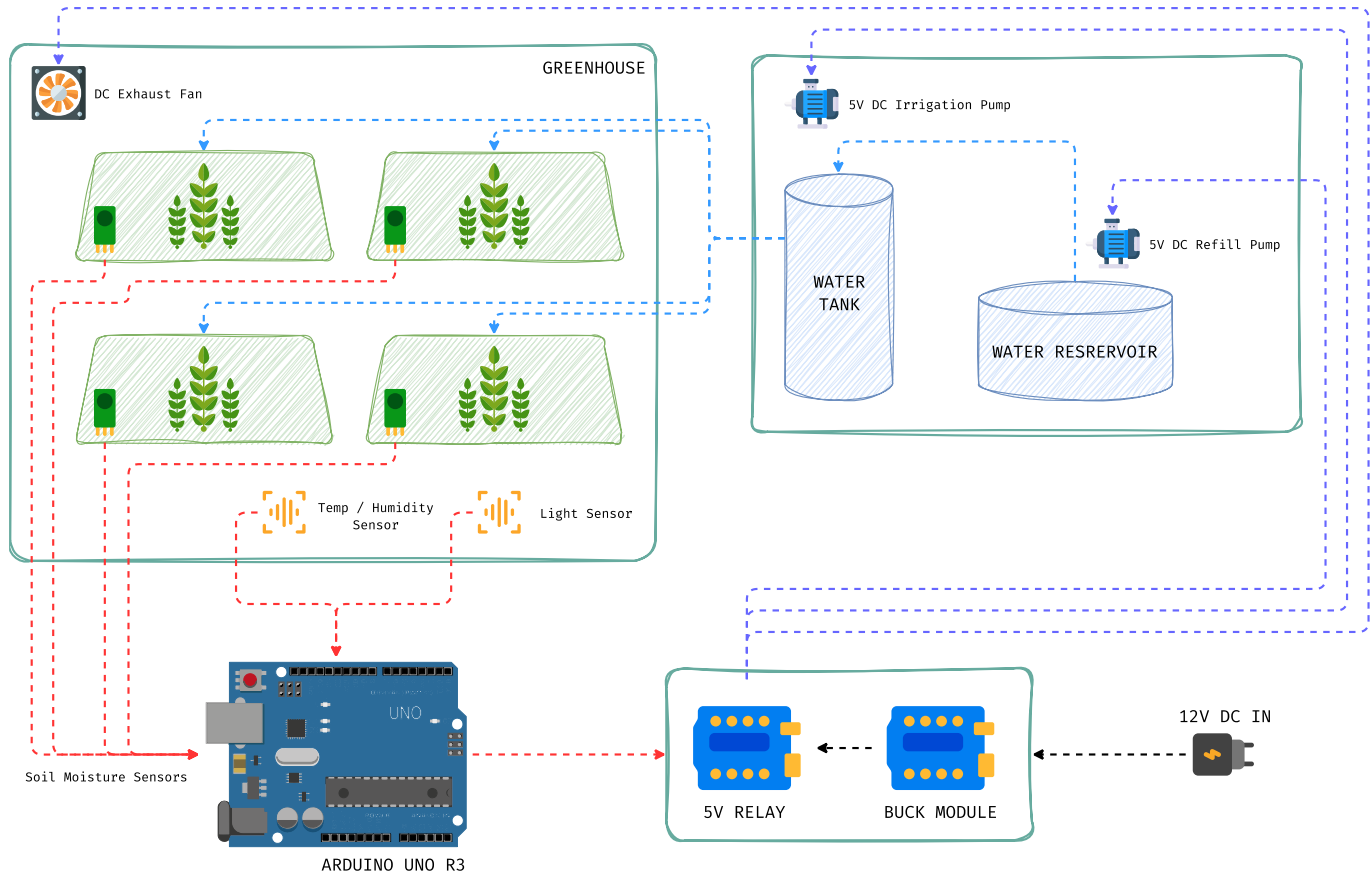


Figure 4 - System Design

Monitoring Dashboard

Clean Autonomous system requires monitoring, because the farmer can only take decisions about the crops once he has the idea about how they are growing. Our custom build system provides all the necessary vital details about the system such as Photosynthesis Level, Temperature & Humidity, Soil Moisture Levels and as a plus point live monitoring of the Water Tank. This set of features covers all the essential monitoring details it requires for a Greenhouse

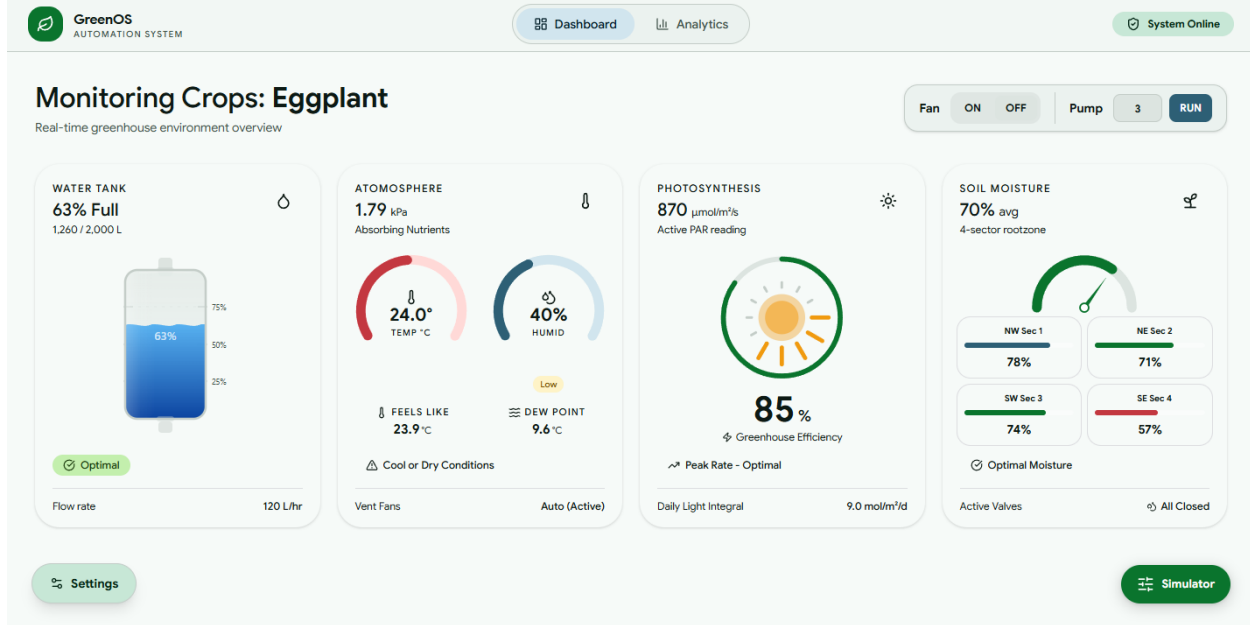


Figure 5 - Monitoring Dashboard

Discussions

While we are developing the system, we had to face many technical difficulties. Unlike software, we must deal with environment when it comes to hardware. Below are the noted areas we find challenging.

Adaptive Signal Smoothing

Acoustic echoes from the water bottle's curved walls caused the HC-SR04 to produce wild telemetry spikes. We bypassed standard Mean Average filters, which are susceptible to outliers, in favor of a Median Filter coupled with an Adaptive Exponential Moving Average (EMA). A standard EMA creates significant "digital lag" during rapid depletion. To solve this, we implemented a 12% Breakaway Threshold. If the change in water level exceeds 12% between polls, the smoothing factor is discarded, allowing the system to "snap" to real-time data when pumps are active

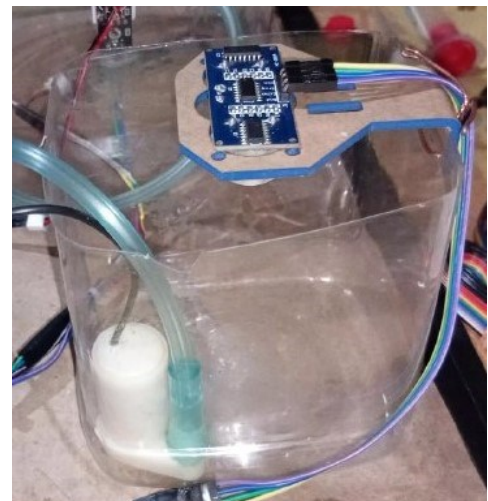


Figure 6 - Water Tank

Photoresistor Module

During the prototyping phase, we utilized a Custom Soldered LDR Module (using a 10k Ω pull-down resistor) as a logic bridge. This allowed us to build out the Analog-to-Digital conversion logic and C# dashboard widgets before the BH1750 I2C sensor was integrated, proving the modularity of the firmware.



Figure 7 - Custom Made LDR Module

Future Implementations

This prototype establishes a functional foundation for autonomous irrigation and refill systems. The architecture is modular, permitting rapid scaling for commercial applications.

- * **IoT Migration:** Utilizing the reserved ESP32 for cloud-native connectivity, moving out the gateway from local USB connectivity.
- * **Modular Hardware Units:** Transitioning from hand-soldered perfboards to dedicated Printed Circuit Boards (PCBs) with integrated flyback protection and screw-terminal connectors.
- * **Commercial Irrigation Expansion:** Replacing the PVC drip manifold with individual solenoid valves to allow for distinct crop-specific nutrient dosing and watering schedules. The C# Gateway's potential as a "Smart Farm Hub" is realized when exposed via Reverse Proxies such as Cloudflare Tunnels. This enables secure, global access to greenhouse telemetry without the security risks of traditional ports.

By combining high-level software intelligence with affordable hardware HALs, we have produced a production-ready solution for the future of small-scale agriculture.